

Mathematical and Computational Methods

Exercise Sheet 1: Foundations: numbers, sets and functions

Tutorial route (►, in this order, about 80 minutes): 1.3(b,d), 2.1, 2.3(b,d), 2.4(b), 2.6(a,b), 2.8(a–c), 2.5. Physics groups: 2.9 in place of 2.8. Everything else is for home study.

Levels: ★ drill, ★★ core, ★★★ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Exercise 1.1 (★ drill — intervals & sets).

- (a) Write $\{x \in \mathbb{R} : -2 \leq x < 5\}$ in interval notation.
- (b) Write $(-\infty, 3]$ in set-builder notation.
- (c) Express $\{x \in \mathbb{R} : x^2 \leq 9\}$ as a single interval.
- (d) Compute $(-1, 4] \cap [2, 7)$.
- (e) Compute $[0, 5] \setminus (2, 3)$.

Exercise 1.2 (★ drill — Σ and Π notation). Evaluate

$$(a) \sum_{k=1}^4 (3k-2), \quad (b) \sum_{k=1}^5 k^2, \quad (c) \prod_{k=1}^4 k, \quad (d) \prod_{k=2}^4 \left(1 - \frac{1}{k}\right).$$

Exercise 1.3 (► ★ drill — inequalities). Solve for real x and give the answer as an interval (or union of intervals).

$$(a) 3 - 2x \leq 7, \quad (b) x^2 - x - 6 > 0, \quad (c) |2x - 1| \leq 3, \quad (d) |x + 2| > 1.$$

Exercise 1.4 (★ drill — maximal domain). State the maximal (natural) domain of each function.

$$(a) f(x) = \frac{1}{x-2}, \quad (b) g(x) = \sqrt{x-4}, \quad (c) h(x) = \ln(x+1), \quad (d) k(x) = \frac{1}{\sqrt{3-x}}.$$

Exercise 1.5 (★ drill — composition). Let $f(x) = 2x + 1$, $g(x) = x^2$ and $k(x) = \frac{1}{x}$.

- (a) Find $(f \circ g)(x)$, $(g \circ f)(x)$ and $(f \circ f)(x)$, each with its domain.
- (b) Evaluate $(f \circ g)(3)$.
- (c) Find $(k \circ k)(x)$ and its domain.

Exercise 1.6 (★ drill — classify, then invert). For each map $f: \mathbb{R} \rightarrow \mathbb{R}$, state whether it is injective, surjective, both (bijective) or neither; for each bijective one, find f^{-1} .

$$(a) f(x) = 3x - 1, \quad (b) f(x) = x^2, \quad (c) f(x) = x^3, \quad (d) f(x) = e^x.$$

(e) Find the inverse of $f(x) = \sqrt{x} + 1$ on $[0, \infty)$, with its domain.

Exercise 1.7 (★ drill — even / odd / periodic). Classify each as even, odd, or neither; for (d) also state the period.

$$(a) x^4 - 3x^2, \quad (b) x^3 + x, \quad (c) x^2 + x, \quad (d) \cos(2x).$$

Exercise 1.8 (★ drill — graph transformations). Describe in words how the graph of $y = f(x)$ moves to give (a) $y = f(x) + 3$, (b) $y = f(x - 2)$, (c) $y = -f(x)$, (d) $y = f(2x)$. Then, starting from $y = x^2$, identify the transformations giving $y = (x + 1)^2 - 4$ and state its vertex.

2 Tutorial problems

Exercise 2.1 (► ★★ core — maximal domains and composites).

- (a) Find the maximal domain of $r(x) = \frac{\sqrt{x+3}}{x-1}$, in set-builder and in interval notation.
- (b) Let $f(x) = \sqrt{x}$ and $g(x) = 1 - x^2$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$, each with its domain.
- (c) Express $h(x) = (2x + 1)^5$ and $p(x) = \frac{1}{\sqrt{x^2 + 1}}$ each as a composition of simpler functions, naming the inner and the outer function.

Exercise 2.2 (★★ core — inequalities, two ways). Solve

$$(a) \frac{2x+1}{x-3} \geq 0, \quad (b) |x-1| < |x+3|.$$

Exercise 2.3 (► ★★ core — x^2 and x^3 : bijectivity and inverses).

- (a) Show that $f(x) = x^2$, as a map $\mathbb{R} \rightarrow \mathbb{R}$, is neither injective nor surjective.
- (b) Show that the same rule, as a map $[0, \infty) \rightarrow [0, \infty)$, is bijective, and write down its inverse.
- (c) Explain why $f(x) = x^3$ is bijective on $\mathbb{R}$, and find f^{-1} .
- (d) Find the inverse of $g(x) = (x - 1)^3 + 2$ and verify that $g(g^{-1}(x)) = x$.

Exercise 2.4 (► ★★ core — graph transformations).

- (a) Obtain $g(x) = (x - 2)^2 + 1$ from $y = x^2$ by a sequence of transformations; state the vertex and sketch both curves.
- (b) Obtain $h(x) = 3 - \sqrt{x+2}$ from $y = \sqrt{x}$. State the transformations used, the domain and the range of h , and whether h is increasing or decreasing.

Exercise 2.5 (► ★★ core — even and odd parts (maths)).

- (a) Prove that the product of two odd functions is even.
- (b) Show that any function f on a domain symmetric about 0 can be written $f = E + O$ with $E(x) = \frac{1}{2}(f(x) + f(-x))$ even and $O(x) = \frac{1}{2}(f(x) - f(-x))$ odd. Apply this to $f(x) = e^x$.

Exercise 2.6 (► ★★ core — reading the piecewise ReLU). Recall $\text{ReLU}(x) = \max(0, x)$.

- (a) Write ReLU as a piecewise formula, and evaluate it at $x = -3, 0, 2$.
- (b) Show that $\text{ReLU}(x) = \frac{1}{2}(x + |x|)$.
- (c) Hence express $|x|$ in terms of $\text{ReLU}(x)$ and $\text{ReLU}(-x)$.

Exercise 2.7 (★★ core — *trigonometric equations*). Solve on $[0, 2\pi)$, giving exact answers in radians.

$$(a) 2 \sin x - 1 = 0, \quad (b) 2 \cos^2 x - 1 = 0, \quad (c) \sin x = \cos x.$$

Exercise 2.8 (► ★★ core — *the sigmoid as a probability (data science)*). The logistic (sigmoid) function is $\sigma(x) = \frac{1}{1 + e^{-x}}$.

- Compute $\sigma(0)$ and explain why $0 < \sigma(x) < 1$ for every real x .
- Verify the symmetry $\sigma(-x) = 1 - \sigma(x)$.
- In logistic regression a real-valued score s is turned into a class probability $\sigma(s)$. If $s = \ln 4$, find the probability. What score gives probability $\frac{1}{2}$?
- Sketch $y = \sigma(x)$ by hand, marking $\sigma(0)$, the two horizontal asymptotes, the direction of monotonicity and the centre of symmetry, each justified in one line (you may use (a) and (b)).

Exercise 2.9 (★★ core — *a potential barrier from steps (physics)*). The rectangular potential barrier of height $V_0 > 0$ and width $2a$ is $V(x) = V_0$ for $-a \leq x < a$ and $V(x) = 0$ otherwise. (Which of the two endpoints carries the value V_0 is physically immaterial; we fix it here so that the step formula below is exact.) Throughout, H is the Heaviside step, with the convention $H(0) = 1$ (so $H(x) = 1$ for $x \geq 0$ and $H(x) = 0$ for $x < 0$).

- What is the range of V ? Where is V discontinuous?
- Show that $V(x) = V_0[H(x+a) - H(x-a)]$, and check the value at $x = 0$ and at $x = 2a$.

3 Further practice (home)

Exercise 3.1 (★★ core — *rational functions: graph, range, inequality*).

- Starting from $y = \frac{1}{x}$, describe the graph of $y = \frac{1}{x-2} + 1$, giving its maximal domain and both asymptotes.
- Find the range of $f(x) = \frac{x^2}{x^2 + 1}$.
- Solve $\frac{x}{x+1} < 2$.

Exercise 3.2 (★★ core — *maximal domains, harder*). State the maximal domain of each, in set-builder and in interval notation.

$$(a) f(x) = \ln(x^2 - 4), \quad (b) g(x) = \frac{\sqrt{x^2 - 1}}{x - 3}, \quad (c) h(x) = \arcsin(x - 1).$$

Exercise 3.3 (★★ core — *exponentials and logarithms: equations and graphs*).

- Solve $3e^{2x} = 12$.
- Solve $\log_2 x + \log_2(x-2) = 3$.
- Solve $e^{2x} - 5e^x + 6 = 0$.
- On one set of axes sketch $y = e^x$, $y = \ln x$ and $y = x$, marking where each curve meets an axis and each asymptote, and explain why the two curves are reflections of each other in $y = x$. Hence sketch $y = \ln(x-1)$, stating its domain and asymptote.

Exercise 3.4 (★★core — *tan, cot and the inverse trigonometric functions*).

- On separate axes, sketch $y = \tan x$ and $y = \cot x$ over $[-\pi, \pi]$. Mark every vertical asymptote and every x -intercept in that range, and state the period of each.
- $\arctan$ is the inverse of $\tan$ restricted to $(-\frac{\pi}{2}, \frac{\pi}{2})$. State its domain and range, explain why $y = \pm\frac{\pi}{2}$ are horizontal asymptotes and why $\arctan$ is odd, evaluate $\arctan 0$ and $\arctan 1$, and sketch $y = \arctan x$.
- State the domain and range of $\arcsin$, $\arccos$ and arccot . Evaluate $\arcsin \frac{1}{2}$, $\arccos(-\frac{1}{2})$, $\operatorname{arccot} 0$ and $\operatorname{arccot} 1$, and give the horizontal asymptotes of $y = \operatorname{arccot} x$.

Exercise 3.5 (★★core — *hyperbolic functions from the definitions*). Work directly from $\sinh x = \frac{1}{2}(e^x - e^{-x})$ and $\cosh x = \frac{1}{2}(e^x + e^{-x})$.

- Evaluate $\sinh 0$, $\cosh 0$ and $\tanh 0$.
- Show that $\sinh$ is odd and $\cosh$ is even.
- Prove $\cosh^2 x - \sinh^2 x = 1$.
- Hence show that $\cosh x \geq 1$ for every real x , and find where equality holds.

Exercise 3.6 (★★core — *the logit (data science)*). For the sigmoid $\sigma(x) = \frac{1}{1 + e^{-x}}$ (range $(0, 1)$):

- Show that its inverse is the *logit* $\sigma^{-1}(p) = \ln \frac{p}{1-p}$, $0 < p < 1$.
- A classifier outputs probability $p = 0.8$. What score s produced it?

Exercise 3.7 (★★★challenge — *the Gauss sum, and using Σ and Π*).

- Prove $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ by writing the sum forwards and backwards and adding, with no appeal to induction.
- Hence show that $\sum_{k=1}^n (3k+2) = \frac{3n^2 + 7n}{2}$, and evaluate it at $n = 10$.
- Evaluate $\prod_{k=1}^n \frac{k+1}{k}$ for general n .

Exercise 3.8 (★★★challenge — *monotonic $\Rightarrow$ injective, but not conversely*).

- Prove that a strictly increasing function is injective.
- Give an injective function on $[0, 2]$ that is *not* monotonic.

Exercise 3.9 (★★★challenge — *irrationality of $\sqrt{3}$*). Prove that $\sqrt{3}$ is irrational, imitating the proof for $\sqrt{2}$ in the notes. (Every integer has the form $3m$, $3m+1$ or $3m+2$.)